

Exercises: Eigenvalues and Eigenvectors

$$A = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \quad B = \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad D = \begin{pmatrix} 4 & 2 \\ 0 & -1 \end{pmatrix}$$

Question 1

Find the eigenvalues for A, C and D .

Answer:

Eigenvalues for Matrix A :

$$\begin{aligned} \det(T - \lambda_A I_2) &= 0 \\ \det \begin{pmatrix} 4 - \lambda_A & 0 \\ 0 & 4 - \lambda_A \end{pmatrix} &= 0 \\ (4 - \lambda_A)(4 - \lambda_A) &= 0 \\ (4 - \lambda_A)^2 &= 0 \\ \lambda_A &= 4 \end{aligned}$$

Eigenvalues for Matrix C :

$$\begin{aligned} \det(T - \lambda_C I_2) &= 0 \\ \det \begin{pmatrix} 1 - \lambda_C & 0 \\ 0 & -1 - \lambda_C \end{pmatrix} &= 0 \\ (1 - \lambda_C)(-1 - \lambda_C) - 0 &= 0 \\ \lambda_C = 1 \text{ or } \lambda_C = -1 \end{aligned}$$

Eigenvalues for Matrix D :

$$\begin{aligned} \det(T - \lambda_D I_2) &= 0 \\ \det \begin{pmatrix} 4 - \lambda_D & 2 \\ 0 & -1 - \lambda_D \end{pmatrix} &= 0 \\ (4 - \lambda_D)(-1 - \lambda_D) - (2)(0) &= 0 \\ (4 - \lambda_D)(-1 - \lambda_D) &= 0 \\ \lambda_D = 4 \text{ or } \lambda_D = -1 \end{aligned}$$

Question 2

Find the eigenvectors corresponding to the eigenvalues for A, C and D .

Answer:

Eigenvectors for Matrix A : We found that there was one eigenvalue for matrix A , $\lambda_A = 4$.

$$\begin{aligned} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= \lambda_A \begin{pmatrix} x \\ y \end{pmatrix} \\ \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} &= 4 \begin{pmatrix} x \\ y \end{pmatrix} \\ 4x + 0y &= 4x \\ 0x + 4y &= 4y \end{aligned}$$

Both equations simplify to $0 = 0$ so any non-zero vectors in $\mathbb{R}^2$ is an eigenvectors of A . The eigenspace is $E_4 = \mathbb{R}^2$ with basis $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ so $\lambda_A = 4$ has geometric multiplicity of 2.

Eigenvectors for Matrix C : We found that there are two eigenvalues for matrix C , $\lambda_C = 1$ and $\lambda_C = -1$.

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (1) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{aligned} x + 0y &= x \\ 0x - y &= y \\ y &= 0 \end{aligned}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (-1) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{aligned} x + 0y &= -x \\ 0x - y &= -y \\ x &= 0 \end{aligned}$$

From the above, when $\lambda_C = 1$ then $y = 0$ with no constraint on x , so $E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$.

When $\lambda_C = -1$ then $x = 0$ with no constraint on y , so $E_{-1} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$. Each of the eigenspaces are one-dimensional, so both eigenvalues have geometric multiplicity 1.

Eigenvectors for Matrix D : We found that there are two eigenvalues for matrix D , $\lambda_D = 4$ and $\lambda_D = -1$.

$$\begin{pmatrix} 4 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (4) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{aligned} 4x + 2y &= 4x \\ 0x - y &= 4y \end{aligned}$$

these simplify to:

$$y = 0$$

$$\begin{pmatrix} 4 & 2 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = (-1) \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{aligned} 4x + 2y &= -x \\ 0x - y &= -y \end{aligned}$$

these simplify to:

$$\begin{aligned} 5x &= -2y \\ y &= -\frac{5}{2}x \end{aligned}$$

From the above, when $\lambda_D = 4$ then $y = 0$ with no constraint on x , so $E_4 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$.

When $\lambda_D = -1$ then $y = -\frac{5}{2}x$, so $E_{-1} = \text{span} \left\{ \begin{pmatrix} 2 \\ -5 \end{pmatrix} \right\}$.

Each of the eigenspaces are one-dimensional, so both eigenvalues have geometric multiplicity 1.

Question 3

Since A is of the form $\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix}$, we know it is a uniform scaling transformation with a scale factor of 4. Using the fact that:

$$B = \begin{pmatrix} 0 & -\frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1/2 & 0 \\ 0 & 1/2 \end{pmatrix} = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$$

, predict what transformation B is.

Answer:

B is a combination of uniform scaling by $\frac{1}{2}$ and then rotation counter-clockwise by 90 degrees. Note the order does not matter here as uniform scaling will commute with everything.

Question 5

Using the fact that the eigenvectors of a linear transformation only change in magnitude under that linear transformation, try to predict what linear transformation C represents. Test out your theory using some simple areas.

Answer:

The eigenspace for C is $E_1 = \text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$ and $E_{-1} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$.

The eigenvectors stay on their own lines so for C they tell us when $\lambda_C = 1$ then all vectors on the x -axis are left alone but when $\lambda_C = -1$ then vectors on the y -axis are flipped to the negative side. This is the definition of a reflection and this is also confirmed by the determinant of C which is -1 so area is

preserved (magnitude of 1) but the orientation is reversed (has a negative sign). The mirror line is E_1 which is the x-axis.

I confirmed this with some simple quadrilaterals using GeoGebra Graphs Tool.